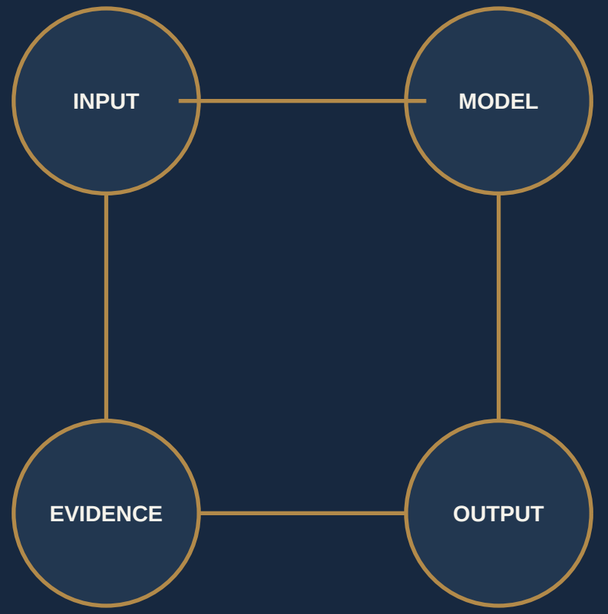
BACHELOR OF ARTIFICIAL INTELLIGENCE & ROBOTICS ENGINEERING

AI Ethics

AI.404 | University lecture series



What you will learn

A practical progression from core ideas to engineering judgment

- Explain and define responsible ai.
- Apply key methods including algorithmic bias.
- Build or analyze a practical system involving fairness.
- Evaluate assumptions, limitations, performance and responsible use.

A connected learning journey

Each stage prepares the tools needed for the next.

01

FOUNDATIONS

Responsible AI

02

METHODS

Algorithmic bias

03

SYSTEMS

Fairness

04

PRACTICE

Laboratory, evaluation and project

The subject at a glance

A map of the ideas and techniques in this course

01 · Responsible AI

Responsible AI

02 · Algorithmic bias

Algorithmic bias

03 · Fairness

Fairness

04 · Explainability

Explainability

05 · Privacy

Privacy

06 · Accountability

Accountability

07 · Governance and societal conse

Governance and societal consequences

08 · Mandatory practical: Conduct a

Mandatory practical: Conduct algorithmic fairness and explainability assessments

Core ideas, made explicit

AI.404 • Distinguish definitions, assumptions and evidence

- Responsible AI
- Algorithmic bias
- Fairness
- Explainability
- Privacy

DEFINE → MODEL → TEST → INTERPRET

Core ideas, made explicit

AI.404 • Distinguish definitions, assumptions and evidence

- Accountability
- Governance and societal consequences
- Mandatory practical: Conduct algorithmic fairness and explainability assessments

DEFINE → MODEL → TEST → INTERPRET

Follow the system through its stages

Use the process diagram to connect components and outcomes.

PROBLEM
Responsible AI

INPUTS
Algorithmic bias

METHOD
Fairness

OUTPUT
Explainability

EVALUATION
Evidence, constraints and limitations

Teaching sequence: reveal one component at a time; then replay the complete end-to-end flow.

From concept to a real engineering decision

A short case study for discussion and worked-example narration

SCENARIO

A real task involving responsible ai.

CONSTRAINTS

Data, resources, safety and operating conditions shape the design.

DECISION

Select a method; justify the trade-offs and define success.

- State assumptions before calculating or implementing.
- Compare the expected result with observed evidence.

Laboratory · build, measure, improve

Students work through the following guided sequence.

- Specify a measurable task using responsible ai.
- Implement or simulate a baseline using algorithmic bias.
- Run a controlled test and record results with units and conditions.
- Diagnose a failure, revise the design and explain the improvement.

Review · explain · apply

KNOWLEDGE CHECK

What assumptions does the method make? What evidence would show that it works?

PRACTICAL CHALLENGE

Apply one course idea to a new context; state constraints, test criteria and likely failure modes.

Key takeaway: make the model visible, test it, and communicate its limits.

Course-series slide design • detailed narration, demonstrations and assessments expand each module.